Machine Learning Algorithm Model for Improving Business Decisions Making in Upstream Oil & Gas

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***Abstract*— The upstream capital project oil and gas industry is considered a critical sector in Malaysia. Apart from its significant monetary contribution to the country, big data analysis is also applied to the supply chain operation. The prescriptive analysis is based on Artificial intelligence (AI), specifically Machine Learning (ML), which involves algorithms and models that enable computers to make decisions based on mathematical data relationships and patterns. This study aims to identify ML analysis in Malaysia's upstream capital projects, which may improve business decisions via the use of statistical models and ML algorithms. Incorporating ML algorithms and statistical models will produce better business decision-making by enhancing efficiency and productivity besides fast monetisation and minimising risk and returns. Overall, with the use of mixed analysis elements, it can produce better decision support for stakeholders and company owners before making crucial business decisions.**

Keywords*— *Algorithm Model, Business Decisions, Upstream, Oil & Gas, Prescriptive Analysis

I. INTRODUCTION

Nowadays, many companies are involved with oil and gas downstream, upstream or integrated upstream and downstream activities. Upstream activities are mainly focusing on exploration and production. Meanwhile, downstream activities focusing on activities like conversion of oil and gas into finished products such as petroleum, natural gas oils, diesel, and other energy supplies. To ensure business's continuity, making the right decision is the key to ensure the success of a company. However, the decision-maker cannot produce the most accurate decision without detailed analysis based on a huge data received. The incorporation of ML and big data is the best approach to expedite the process of decision-making. The prescriptive analysis uses statistical models and ML algorithms to define the possibilities and to recommend actions. These models and algorithms can discover patterns in big data that human analysts may overlook. The prescriptive analysis relies on AI, particularly in the field of ML, which include algorithms and models that allow computers to make decisions based on the relationships and patterns of statistical data.

II. MAIN BUSINESS OBJECTIVE

In order to formulate strategic planning in decision-making and maintain an efficient organisation, a strategy analysis is an essential step to be taken. The organisation's priorities or targets are accomplished with the aid of action strategies. Organisations must perform daily strategy planning for them to progress. This activity will allow them to identify and evaluate critical areas that need modification, predict what would happen, and come with an actionable plan. It is important to determine how substantive progress can be introduced in order for an organisation to perform effectively [1].

Large, centralised organisations will stifle profitable operations and global economic development in the long run. The modern economy will remain intangible-focused to capitalise on the benefits of technical productivity, highly skilled labour, and cutting-edge imagination. The significance of monetisation is widely acknowledged. Systemic study in both logical and scientific forms is virtually non-existent. One of the most important properties of the degree and course of economic growth is the monetisation ratio, which is a component of the economy's goods and services. A lot of these early monetisation efforts have been started based on recent performance.

In today's business paradigm, monetisation sources had earlier been the key to competitive advantage. This requires generating faster, cost-effective development plans through economic analysis of efficient technologies that quickly followed by implementing project management [2]. Each work must be done to ensure that the information required to follow the purpose. Any merging, consolidation, addition, analysis, cleaning, identification, packaging, access, and maintenance must provide the information that meets the requirements. This is a data management mechanism commonly used by many organisations, which is now used for monetisation purposes, as shown in Fig. 1.

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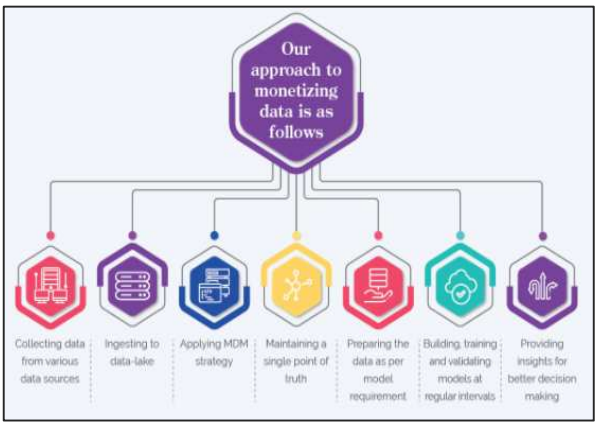

Fig. 1. The framework of Monetizing Strategic [2]

Although with the increasing awareness of the need to deliver more analytical information than generating profit, data and analytics leaders seldom have the knowledge or experience to apply these ideas in their businesses. Lack of accurate and current internal and external data inventory is part of information asset management practices that prevent monetisation from being achieved. Internal politics also hinder innovation; while the department continues to protect its data, the chance to create an enterprise-level monetisation strategy cannot be realised. To capitalise on potential information monetisation offers, business leaders should establish information monetisation functions, develop and maintain an inventory of information assets that may contribute to monetisation, and consider direct and indirect opportunities for monetisation by looking outside their industry organisations.

III. TWO MAJOR SITUATIONS

From many scenarios that usually happen during the process of decision-making, two significant situations are highlighted in this study:

A. Managing Uncertainty for New Investment

Several factors need to be considered before deciding to start a new business or investment. Firstly, the stakeholder will review the assessment of the development concept through a financial roadmap. The decision-maker should achieve financial stability and reap the rewards of money management over time if they use the financial blueprint and follow through with an intelligent approach. Secondly, the subject matter expert will explore the due diligence on the project economics to ensure the significant value of buying a business both in time and money, get detailed information about the quality of financial information and brevity on the information provided in the negotiation phase of the business transaction.

Besides that, consideration of an appropriate factor contributed to risk and value. Fig. 2 shows how a stakeholder considering the risk and value before determining the decision and action [3]. The traditional approach takes a long time, and a huge amount of money is needed in order to generate an accurate feasibility study during the new business selection. This approach requires big teams of economists, planners, project managers without considering IT and statistic experts.

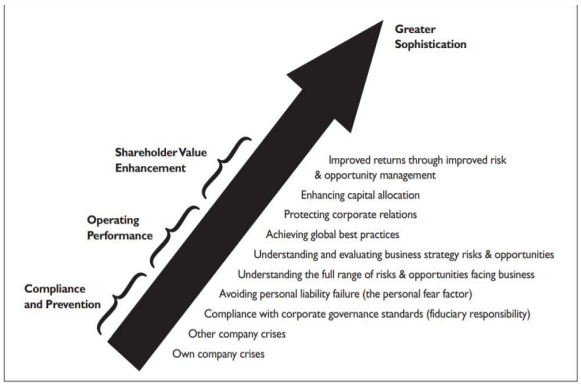

Fig. 2. Evaluation of Risk and Value from the Stakeholder Perspective [9]

However, in the modern era of digital technology, most industrial players utilise big data analysis to improve their analytical method in order to produce the best result to be commenced for their new business. ML is among the top solution in big data analysis. The decision-makers can easily select any new business or investment that could be profitable based on the analytical investigation that is effective in terms of time and cost.

B. Business Strategic Manoeuvre Due to a Recent Issue

A valuable lesson can be learned from the previous economic crisis for over ten years since 2008. The financial crisis sent the world into the Great Recession, which was the most significant economic downturn since the Great Depression. Furthermore, businesses around the globe are facing the devastating impact of the COVID-19 pandemic today. This is the time for industry players to re-strategise on the way they manage their business, thus making it sustainable and crisis-resistant. After the 2008financial crisis, two working ideas emerged, one of which harkens back to Keynes. These turbulent times demand bolder actions. Retrenchment, large government stimulus expenditures, and unorthodox asset purchases by the central bank were all required. The other explanation is that the financial crisis was avoided, and life returned to normal.

Project owners need to devise strategic decision while facing the recent issue to accelerate progress and discover other opportunities to facilitate the process. Today, many big companies are equipped with a dashboard approach that enables them to monitor their project performance. This tool provides real-time results in the form of graphs, charts, etc. Despite that, there is a lack of solution when the project involves many disciplines, a huge workforce and machinery, and different locations and complexity [4]. In short, it is still within the scope of big data. Another alternative when discussing about cost optimisation due to high impact risk during a crisis like COVID-19 pandemic is the introduction of the working from home (WFH) policy. By which the company can save more than 30% of their monthly operation cost.

With the initiative of WFH policy, the company can still proceed with the business, as usual, thus ensuring their survival during a crisis. Different fields indeed come with a different approach. The oil and gas industries involved with upstream activity have to put their projects on hold. This calls them to revisit their capital investment plans. With cash flow forecast in mind, they need to take the necessary step during the near term rather than focusing on running the business as usual. Most of them are focusing on inventory management

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while freezing the operation at the site. They can reduce the cost to pay the wages while the business is put on hold.

IV. Attributes Which Influence Decision Patterns

Functional specifications, configuration constraints, and consistency attribute requirements are examples of input-driven attributes that system members prioritise based on market and project goals. When the platform is used in specific cases, functional specifications define what functions the technology must have to satisfy the specified and inferred stakeholder needs. Several factors influence decision-making, such as project economics, project complexity, risk, fiscal term, geographical, country political, and hydrocarbon price influence. The importance of data in revenue lies in its consistency and continued growth. This enables companies to create new business opportunities, generate more revenue, predict future trends, optimise current business operations, and generate actionable insights.

A. Project Economics

Prime ideas that currently drive the global innovation economy are different from the industrial economies of the past. An innovative economy alone cannot produce vitality and quality of life in an area if the population does not have the skills to participate in the growth of high-level employment opportunities. However, in the long run, one cannot maintain vitality and quality of life without being involved in an innovative economy.

The success of the project refers to the solutions and a better strategy for implementing various processes. Success like that would improve the living conditions and improve methods of doing things. The project's economic decisions are made daily from the individual level to the central banks of various countries [5]. Globalisation has made markets more interdependent and difficult to evaluate. The input model is primarily used in the economic assessment to forecast possible economic movements. Only an accurate interpretation of economic factors will lead to effective economic decisions.

B. Project Complexity

The decision of which resources will be used to achieve the project objectives in the most efficient way depends on the particular project's characteristics, such as its objectives, resources, and context. There are various predictive techniques available, ranging from basic linear regression to sophisticated, efficient ones such as artificial neural networks. Complex models normally do well in terms of inference, but they are elusive and unable to clarify forecasts or patterns found. [6]. According to Shtub et al. [6], complexity is the root of the project concept. The factors determining a project's implementation include unprecedented complexity in its design, development, and implementation. Remington et al. [7] agree that complexity is vital for project management because of decision-making and goal achievement difficulties. However, they show a lack of operational definition for complex projects.

C. Risk & Influence

Risk is a necessary component of decision-making. Furthermore, risk perception may be seen as an individual risk assessment, and the accuracy of risk assessment is dependent on the availability of accurate risk data. Several factors identified as influencing risk perceptions associated with risk design are framing effects, trust and confidence sources, and target risk. A review of Kusev et al. [8] shows that decision-making does not depend on the context. In particular, the difference in contextual risk descriptions, such as the amount of money and accessibility to clean content, make decisions about the events in memory and emotion are activated to change the selection of risks [9].

D. Fiscal Term

The extractive industry expands and works in most countries through partnerships between sovereign governments and private corporations. The fiscal factors that control these parties' partnership specify how the financial gains and costs of the extractive project can be shared. Terms can be found in country regulations that extend to all programmes or in-person contracts that regulate the website's rights and responsibilities of a single project.

Fiscal terms in the oil, gas, and mining sectors must be organised around four main characteristics of the extractive sector [10]: (i). Petroleum and mineral resources are not limited. Therefore the government must generate sufficient returns to balance the country with the value of its depleted assets; (ii). Extractive projects require significant initial investment before income begins to flow; (iii). Geological risks, market fluctuations, technological uncertainty, and political risks are all common project risks.; and (iv). Extractive income has the potential to represent the dominant portion of a country's public income.

E. Geographical

Geography is the study of locations and people's interactions with their surroundings. Geographers study the physical characteristics of the earth's crust and the human civilisation that exists there. They investigate how human society communicates with the natural world and how their surroundings influence humans. For additional translation material, further information about this source text is needed. Some people struggle to grasp the full breadth of geography because, unlike other disciplines, geography is not defined by a single subject as it deals with many issues and variations related to culture, politics, placement, plants, landforms, and many more.

F. Political climate

In a world of conflict, examining national political stability perceptions is an important metric to consider prior the formulation of social and economic policies. Political stability assists world leaders, companies and organisations to determine the amount of investment they should pour into a certain country. For developed and affluent countries, political stability is given utmost importance as; theoretically, political stability may provide a more stable business climate. Businesses must decide whether a country trusts in free trade, regulatory oversight, or government interference; hence the investment is worthy, and the risk can be minimised.

The country's view on capitalism is also another factor to be considered in business operations. In a broad sense, capitalism is an economic system in which production tools are privately owned and operated. On the other hand, a centralised economy is one in which the government or state directs and controls the economy, including the means of development and decision-making [11]. Historically, democratic governments have favoured liberalism, while totalitarian regimes have favoured state-run economic policies.

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G. Hydrocarbon Price

Hydrocarbons are naturally occurring chemicals that make up crude oil, natural gas, coal, and other important energy sources. As hydrocarbons are being burnt, they contain carbon dioxide, water, and fire, among other things. As a result, hydrocarbons are a very efficient fuel source. The market price of hydrocarbons takes into account device segments and regions, market share, and market size. Besides that, the forecasting for each component of the production and application for the hydrocarbons has been made available to the worldwide market by the industry's expert. From the forecasting of the hydrocarbons price, we can get an understanding of the new product market analysis, financial overview, strategies, and marketing trends [12]. The price of liquid hydrocarbon gas is strongly associated with the price of crude oil and natural gas, and it follows the crude oil and natural gas price trend [13].

V. Type Of Analysis For Decision Support

Continuation of factors influences decision pattern, and the researcher found that several elements become the baseline or support for the decision made during the analysis process. Each analysis chosen might address a specific problem that contributed to and affected any company profitability's primary goal. Fig. 3 shows few analyses that can be considered while making the decision. The fundamental of the investigation to uncover potentially useful data trends, forecast the results of a new project or improve the value of operating an ongoing project. There are various predictive techniques available, ranging from basic linear regression to sophisticated, efficient ones such as artificial neural networks. Complex models normally do well in inference, but they are elusive and unable to clarify forecasts or patterns found [14]. The design choice of which predictive technique is used becomes even more complex since no method outperforms others over a large set of problems.

It is even difficult to find the best parameter values for a specific technique since these settings also are problem-dependent. However, a general theory in decision support can be used as the comparison between descriptive and prescriptive approaches, like the use of prescriptive theory. Both methods suggest the best solution or most accurate way to make a decision [15]. A few analyses must blend seamlessly, thus becoming a solid foundation which can be the based on supporting the process of decision-making with accurate result.

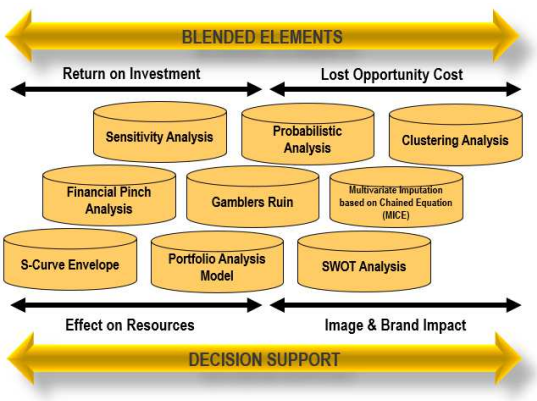

Fig. 3. Type of Analysis for Decision Support [14]

According to statistics, digitalisation is the latest technology, which can be an effective technique in the analysis process leveraging ML and advanced analytics, therefore may add market value in the upstream oil and gas sector. The majority of the world's large corporations aspire to adopt these emerging innovations while still struggling to set their models and products in place, yield meaningful outcomes, and achieve favourable returns on investment.

VI. Conclusion

Analysis of ML algorithm, which is also known as computational performance, is a part of the statistic discipline. With the use of an algorithm, the computer will learn how to match the data with certain information and produce the pattern or trend which will be used to arrive at the best decision [16]. Prescriptive analytics ML in Malaysia Upstream Capital project is new. There are still more rooms and space that can be explored for further improvement. With the two significant scenarios which have been discussed above as references, the researcher has emphasised that the driven attributes decision pattern from project economics, project complexity, risk, fiscal term, geographical, roles of politician and hydrocarbon price may contribute to the analysis pattern during the process of coming up with a strategic decision. A few types of analysis are also supported by the S-Curve Envelope, Financial Pinch Sensitivity, Portfolio, Probabilistic, SWOT, Clustering analysis, and Multivariate imputation based on Chained Equation (MICE). Using a blended or mixed element of analysis can become the better decision support for the company's stakeholder and owner prior arriving at the important decision about the business.

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